

Technical Brief: Gamifying our Eco-Housing solution for LONDON!

Client: homefolk Housing UK CIC (Reg. No. 14593185)

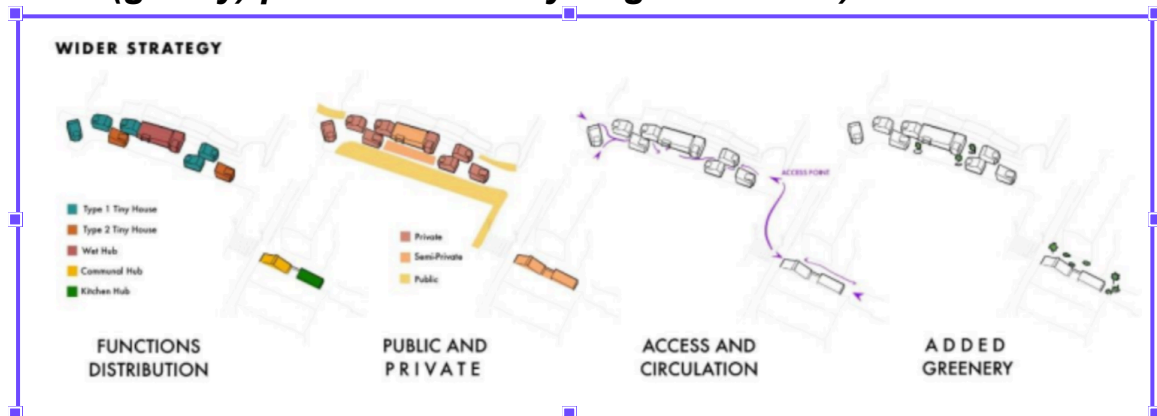
Project: Interactive Digital Resource for Housing Pilot Proposal

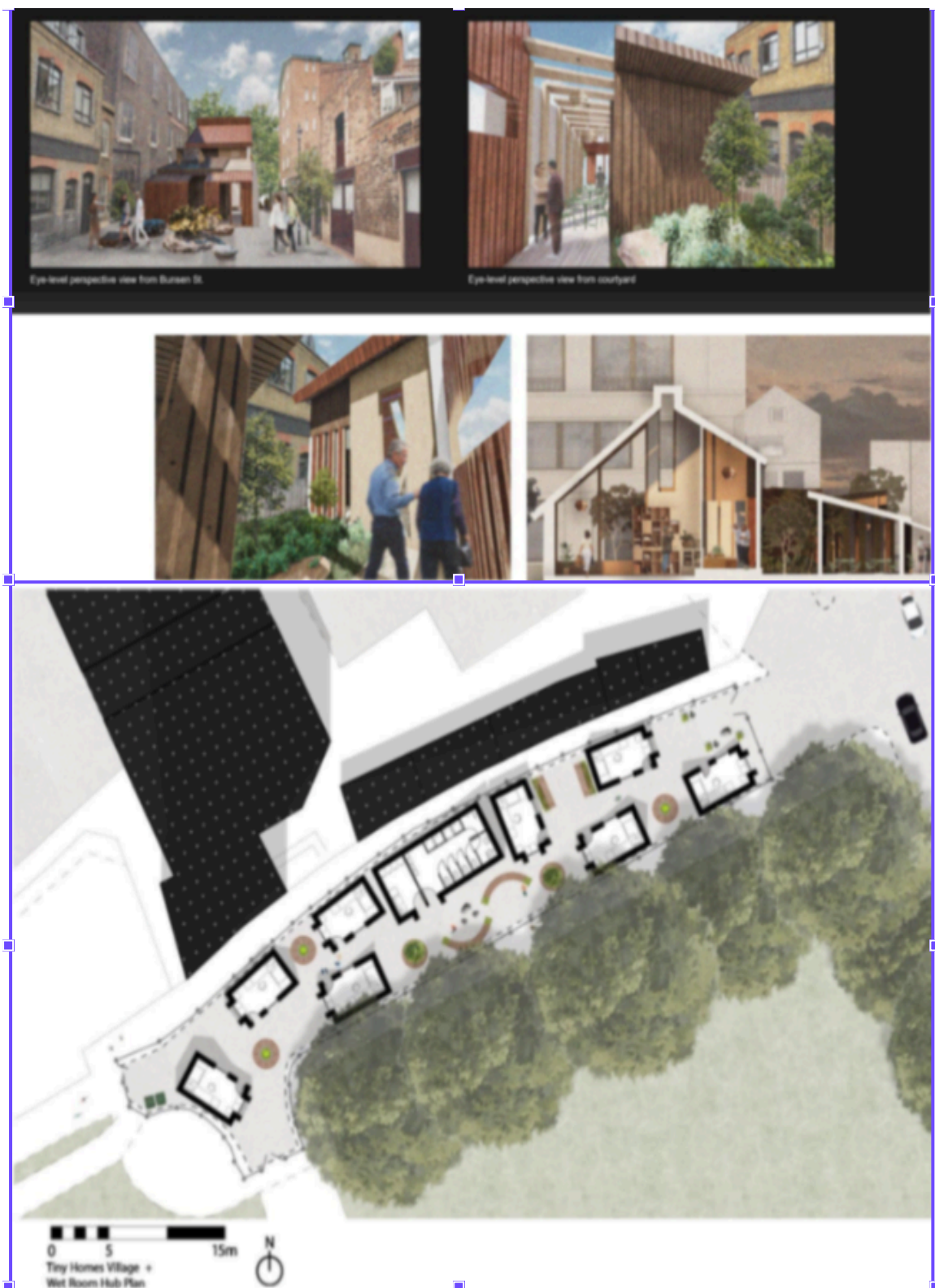
Dates: 1st-4th June 2026

Duration: 10–20 Hours (Hackathon Intensive)

Document Version: 2.0

Nice (grainy) pictures before you get started :)





1. Executive Summary

homefolk is a London-based social enterprise tackling the housing crisis through community-owned tiny-home villages. We seek a **gamified web-based resource** to educate

- academic partners
- potential residents
- and community stakeholders

We want all of the above parties to be able to understand better our housing model. The tool must draw on complex data and previous, disconnected work outputs (legal frameworks, cost structures, site strategy tools & carbon footprints) into an accessible, narrative-driven experience.

Ideally, the solution must prioritise. **Green Coding**. Students must demonstrate awareness of how digital infrastructure contributes to the climate crisis (internet emissions now exceed aviation) and implement strategies to minimise the carbon footprint of their deliverable digital output.

2. Problem Statement

The Dual Crisis

1. **Housing Crisis:** Homes are increasingly **expensive** relative to wages through speculation/financialisation. ~50% of new homes are built by just seven large building companies in the UK! The UK property market is increasingly monopolistic, prioritising profit over mass affordable housing. A new house in London is ~£700k on average! Can you afford that?
2. **Climate Crisis & Digital:** The internet's greenhouse gas output has eclipsed aviation. Software now dumps more carbon into the atmosphere than airplanes! Every line of code, every image loaded, every video streamed contributes to this. The human cost? In 2020, climate disasters forced 30 million people to flee their homes – three times more than those displaced by war and violence. By 2050, the UN estimates, with current climate projects, **over 1.2 billion people will be displaced** by increasing climate disasters globally - 1 in 8 people on the planet.

Global increases in migration, from rural to urban and between regions, for food and safety is a theme that **every young person alive today** should think about when assessing the severity of the ongoing crisis. Our generation **will** create thousands of our own new big ideas (like [ecosia](#), the [fossil fuel non-proliferation treaty](#), or [London Renters Union](#)!). People alive today **will** successfully create new models for joyful, affordable and climate-safe housing & ways of life that address our issues. The biggest barrier is not tools, or resources - collectively, we have more resources and materials now than at any time in human history! The biggest place we need to make a difference? Unlocking imagination.

The Challenge

Create a digital tool that effectively communicates **the homefolk model** without contributing to the digital carbon load. Students must build something that **embodies the values** it promotes — a low-carbon, community-focused solution to the housing crisis.

Key Data Points to Visualize

Metric	Value	Source
Target cost per resident (2023 pricing)	£50,000	homefolk feasibility st
Typical London home cost	£700,000	Market comparison
Potential homes in Hackney	~20,000	Using 5% of borough'
UN affordable housing threshold	<30% of income	UN definition
Digital emissions vs. aviation	Internet > Aviation	2025 sustainability re

3. Technical Requirements & Constraints

A. Functional Scope

- **Format:** Single-page application (SPA) or lightweight static site
- **Interactivity:** *Gamified* elements (resource allocation simulators, branching narrative paths for different stakeholder personas)
- **Content Integration:** Must visualize data points regarding:
 - Cost comparison (£50k vs £700k)
 - Legal frameworks (CIC, CBS, CLT, MHOS — the "homefolk sandwich" model)
 - Carbon footprint of the housing model
 - Site strategy (marginal paved commons, greyfield land)
- **Output:** A deployable prototype ready for presentation on Day 4 of Innovation Fest

B. Green Coding & Sustainability Standards (2025-2026 Best Practices)

We ask students to explore **Sustainable Web Design** principles. The codebase should be optimised for energy efficiency.

Recommendation	Standard	Tool/Method
Carbon Budget	<0.1g CO2e per page view	Website Carbon Calc
Image Formats	WebP/AVIF only	Squoosh-Web UI (CL ImageOptim, TinyPNG)
JavaScript	Minimal bundle size	esbuild, Terser, tree-s
CSS	Compressed	CSS-Nano
Hosting	Green Web Foundation certified	GreenGeeks, Hetzne sustainability regions
Caching	Aggressive	HTTP/2 + HTTP/3, C
Loading	Lazy-load off-screen resources	Intersection Observer

CI/CD Integration	Carbon monitoring plugins	Lighthouse EcoIndex GreenFrame, Carbon
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Energy-Aware Coding Guidelines:

- Avoid tight polling loops
- Prefer efficient algorithms
- Profile code with Energy-Profiler
- Schedule intensive tasks during off-peak, low-carbon grid periods
- Use declarative UI libraries that minimize re-renders

C. Ethical AI Coding for Beginners: Using Claude & Open Tools

For students new to coding, AI assistants like **Claude**, **GitHub Copilot**, and open-source agents can accelerate learning. However, using them ethically requires protecting your code, respecting privacy, and understanding the environmental cost of AI itself.

Why "Ethical" Matters in AI Coding

1. **Data Privacy:** Uploading proprietary code or personal data to public AI models can leak sensitive information.
2. **Environmental Cost:** Running large AI models consumes significant energy. Every prompt you send adds to the digital carbon footprint.
3. **Bias & Accuracy:** AI can hallucinate (invent facts) or suggest insecure code. You must verify every line.
4. **Ownership:** Ensure the code generated respects open-source licenses and doesn't violate copyright.

Recommended Ethical AI Tools (2026)

Tool	Type	Why It's Ethical/Green	Best For
Claude Code	CLI Agent	Anthropic's official CLI; allows local file access with explicit permission; strong privacy controls.	Beginners, command-line, respects f
Aider	Open Source CLI	Model-agnostic (bring your own API key); runs locally or via secure API; edits multiple files simultaneously.	Multi-file r complex p
Cline	VS Code Extension	Open source; asks permission before executing commands or accessing files; supports local models (Ollama).	Interactive Code with
Continue	VS Code/JetBrains Extension	Open source; connects to local or self-hosted models; keeps data on your machine.	Privacy-fo popular IDE

Ollama	Local Model Runner	Run open-source models (like Llama 3, CodeLlama) entirely offline on your laptop.	Maximum leaves yo
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Step-by-Step: Using Claude Safely for Beginners

If you choose to use **Claude** (via the web interface or CLI), follow these steps to stay safe and ethical:

1. Configure Privacy Settings

- **Web Interface:** Go to Settings > Data Controls. Disable "Improve the Model" if you want to prevent your prompts from being used for training.
- **CLI (Claude Code):** Use the `--no-remote` flag if available, or ensure you are only sending code snippets, not entire repositories.
- **Rule of Thumb:** Never paste API keys, passwords, or personal user data into an AI prompt.

2. Craft Secure Prompts (OpenSSF Guidelines)

The **Open Source Security Foundation (OpenSSF)** recommends specific prompting strategies to avoid leaks:

- **Context Limiting:** Instead of uploading the whole file, paste only the relevant function.
 - *Bad:* "Here is my entire `config.js` file..."
 - *Good:* "Here is a snippet of a configuration function. How do I optimize it for lower memory usage?"
- **Sanitization:** Replace real names, IDs, or URLs with placeholders (e.g., `[USER_ID]`, `[API_KEY]`).
- **Verification:** Always ask the AI to explain *why* it suggested a change. "Explain the security implications of this code."

3. Verify Code Before Running

AI can generate code that looks correct but has vulnerabilities (e.g., SQL injection, hardcoded secrets).

- **Run Security Scans:** Use tools like **Grype** or **GitHub Advanced Security** to scan AI-generated code.
- **Manual Review:** Read every line. If you don't understand it, ask the AI: "Explain this line by line."
- **Test Locally:** Run the code in a sandbox environment before deploying.

4. Reduce AI Carbon Footprint

- **Batch Prompts:** Instead of asking many small questions, combine them into one detailed prompt.
- **Use Smaller Models:** For simple tasks (syntax checking), use smaller, faster models (e.g., CodeLlama-7B via Ollama) instead of massive models.
- **Local First:** If you have a capable laptop, run models locally with **Ollama** to avoid sending data to distant data centers.

Beginner Workflow Example: Building a Homefolk Feature

Goal: Create a simple "Cost Calculator" for the homefolk housing model.

1. **Setup:** Install **VS Code** and the **Cline** extension (or use **Claude Code** in terminal).
2. **Prompt:** "I need a simple JavaScript function that calculates housing costs based on a £50,000 base and a 30% income threshold. Keep the code minimal and efficient."
3. **Review:** The AI generates the code. You check it:
 - *Is it efficient?* (Yes, no unnecessary loops).
 - *Is it secure?* (No user input is being sent to an external API).
4. **Refine:** "Add a comment explaining the math and convert the output to a string for display."
5. **Test:** Run the function in your browser console.
6. **Optimize:** Ask, "How can I make this function use less memory?" (AI suggests using `const` instead of `let` and avoiding global variables).
7. **Deploy:** Host the result on a green host (e.g., Netlify).

Resources for Further Learning

- [OpenSSF Security Guide for AI](#): Official guidelines on secure prompting.
- [Coursera: Privacy-Conscious Development](#): Course on using AI safely.
- [Aider Documentation](#): Guide to using the open-source CLI agent.
- [Ollama Models](#): List of open-source models you can run locally.

Final Note for Students

Using AI is a powerful way to accelerate your learning about coding, but **you are the architect**. The AI is just a tool. Always:

1. **Understand** the code you generate.
2. **Verify** it for security and accuracy.
3. **Optimize** it for energy efficiency.
4. **Credit** the tools you use in your sustainability report.

4. homefolk Resources & Document Repository

The following documents are available from homefolk's official website and should be referenced in student prototypes:

Primary Documents

Document	URL	Description
Document Repository	https://homefolk.org.uk/doc-repository	Central hub for all tech
Model Overview	https://homefolk.org.uk/homefolks-model	Core explanation of the model
Tiny Home FAQs	https://homefolk.org.uk/tinyhome-faqs	Definitions and common information
Main Website	https://homefolk.org.uk	General information and

Specific Documents in Repository

1. **ASCA Design Collaboration** (2024) — Graphics and site-specific modelling for 3 East London sites
 - a. 3 architectural concept docs made for a collaboration homefolk was part of https://drive.google.com/drive/folders/1KqWtfvCMI9frwH1Apu54nj_cB3keJ1c4?usp=share_link
2. **Legal Ownership Model: Homefolk Sandwich** (10 pages, Draft 2025) — Layered legal framework to ensure perpetual affordability...
 - A. Extra! An early draft of what's being worked on at Cambridge Uni for homefolk ... TBC
3. **Site Strategy** (13 pages, 2023) — Investigation into scaling using paved surface area studies
 - A. Extra! The result of a previous 1 day hackathon with UCL :) https://drive.google.com/drive/folders/11UNev8N2VZ7sLIIOQDv4Z8o16X_KLAcn?usp=share_link
4. **Financial Planning Document** (16 pages, 2023) — OLD financial strategy (numbers not current but useful for context)
5. **Village Concept Document** — Key technical considerations and project context
6. **Critical/Community Review Document** — Positive and negative perspectives from early consultation
7. **Graphic Timeline** — Historical context of UK housing and homefolk's development pathway
8. **Community Engagement Workshops** — Walking tours, design processes, workshop materials
9. **Social Enterprise Slide Deck** (10 pages) — Overview for fundraising and partnerships

Additional Resources

- MORE on our site strategy!
- Some case studies of OTHER sites that already exist to click though <https://www.canva.com/design/DAGzINSh51M/PxtMZIP5MNugQa4lrYEkdA/edit>
- **Collaborators Page:** Lists academic and NGO partners (UCL, University of Glasgow, University of Cambridge, Sheffield)

Important Note: Some documents are marked "OLD" and contain outdated financial figures. Don't shy away from needing to redo calculations, however, students should clearly label any simulated data.

5. Recommended Open Source Tools & Resources (2025-2026)

A. Development Frameworks (Low Energy Consumption)

Tool	Why It's Green	Best Use Case
Astro	Ships zero JavaScript by default; island architecture	Content-heavy sites v interactivity

Svelte	Compiler moves work to build time; tiny runtime bundles	Interactive application overhead
Hugo	Written in Go; extremely fast build times	Static documentation generation
Eleventy (11ty)	Simple, flexible static site generator	Lightweight content s

B. Gamification & Interaction Libraries

Tool	Why It's Green	Best Use Case
Phaser 3	Open-source HTML5 game framework; optimized rendering	Village simulation, res mini-games
Three.js	WebGL 3D library (use low-poly models)	3D visualization of ho
GSAP	Animation library with efficient rendering	Smooth transitions ar animations
Storytelling.js	Narrative branching libraries	Stakeholder persona

C. Data Visualization

Tool	Why It's Green	Best Use Case
Observable Plot	Lightweight statistical graphics	Housing statistics, co
Chart.js	Canvas-based, optimized rendering	Simple charts for fina
D3.js	Highly customizable (optimize carefully)	Complex data relation

D. Carbon Measurement & Optimization Tools (2025-2026)

Tool	Purpose	URL
Website Carbon Calculator	Measure CO2e per page view	Essential for final sub
Ecograder	Accessibility + sustainability feedback	Pre-submission audit
GreenFrame	Code execution carbon intensity	Development phase r
Carbonalyzer	Browser extension for real-time monitoring	Continuous developm
Calibre	Performance + carbon tracking	CI/CD pipeline integr
Lighthouse EcoIndex	Google Lighthouse sustainability plugin	Automated testing

E. Image & Asset Optimization

Tool	Purpose	Status
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Squoosh-Web UI	Image compression (WebP/AVIF)	Use web version (CLI)
ImageOptim	Lossless compression	Desktop application
TinyPNG	Smart compression	Web service
SVGO	SVG optimization	Command-line tool

F. Hosting & Infrastructure

Provider	Green Certification	Notes
Netlify	Carbon neutral	Generous free tier
Vercel	Carbon neutral	Edge functions for low
Cloudflare Pages	Renewable energy commitments	Global edge network
GreenGeeks	300% renewable energy	Traditional hosting op
Hetzner "Green" Tier	Verified renewable	European data center

Verification: Always check the [Green Web Foundation Directory](#) before selecting a host.

6. Strategic Context for Students

The Narrative Imperative

The tool must not just *tell* the story of homefolk; it must *embody* the values. We want to gamify the issues led to our housing solution as well as our solution itself. But we can't do it all! So we recognise your group are going to take this brief and completely rework it into what showcases your talents best - if that graphics, artwork or heavy stats, that's okay! :)

Note:

Some documents in the repository are marked "OLD" with outdated financial figures. Ideally, please:

1. Use the most recent legal and concept documents as primary sources
2. Clearly label any simulated data as "demonstration purposes"
3. Source stats with publicly available UK housing data (ONS, Shelter, Ministry of Housing) where possible
4. Include disclaimers where data is estimated or outdated

Legal Framework

The "homefolk sandwich" legal model (CIC, CBS, CLT, MHOS, cooperative ownership) is critical to the project but still a work in progress. Misrepresenting these frameworks could undermine the credibility of the pilot proposal. Students should:

- Reference the 10-page Draft Legal Ownership Model (2025)
- Understand the difference between pilot phase (social rent vs 'acting like uni accomodation') and scaled phase (equity accumulation)
- Not shy away from trying to represent the layered ownership structure/alternative to being a renter or classic homeowner. Many people are curious about this.

Climate Consciousness Integration

- **Educate users:** Consider adding a "carbon footprint" indicator to the gamified experience itself? :)

7. Deliverables Checklist

Item	Description
Functional Prototype	Deployed URL
Source Code	GitHub/GitLab repository link
Sustainability Report	PDF or README section with:
— Carbon footprint estimate	Website Carbon Calculator screenshot
Pitch Deck	20-minute presentation
— User journey explanation	How stakeholders interact with the tool
— Technical choices rationale	Why specific frameworks/tools were selected
— Carbon impact analysis	Environmental considerations

8. Evaluation Criteria

Criterion	Weight	Description
Functionality	20%	Does the tool work as intended for free?
User Experience	20%	Is it intuitive, silly, fun and engaging, or is it a simple/accessible n
Content Accuracy	20%	Does it accurately represent the legal model?
Sustainability	20%	Carbon footprint, green hosting
Creativity	20%	Innovative approaches to legal storytelling

9. Next Steps

1. **Verify Homefolk Assets:** Contact organisers if additional documents are needed beyond the public repository. Available on email "admin@homefolk.org.uk" or on 'signal' messaging app username @MapleLea.01 https://signal.me/#eu/-vXcTptle4Q8NHAuE_F2xhxMMrvfZh6c0TuHGrxGDeSpZtJv9INFop8K4zolumMz
 2. **Understand Your Team:** Identify developers, designers, storytellers, and sustainability specialists - think about how you will communicate (online?????? on -->[signal](#)<-- ;)) and check in, who will keep track of 'to dos' and how often do different people need to meet about this?
 3. **Start Small:** Build a "**Minimum** Viable Game" (MVG) first, then layer in complexity
 4. **Measure Early:** Run carbon audits throughout development, not just at the end
 5. **Document for Handover:** Keep notes on why specific technical decisions were made to help the next people pick up where you left off after this hackathon to make your work have a real world impact.
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10. Once again, links (just in case you got exhausted reading all the above)... well done, you did it! :D

- **homefolk Website:** <https://homefolk.org.uk>
- **Document Repository:** <https://homefolk.org.uk/doc-repository>
- **Model Overview:** <https://homefolk.org.uk/homefolks-model>
- **FAQs:** <https://homefolk.org.uk/tinyhome-faqs>
- **Innovation Fest Organizers:** Contact via UCL SU Volunteering (<https://studentsunionucl.org/volunteering/organisation/homefolk>)

11. If you get *boreddddd* and need inspiration...

https://www.bestgames.com/Pou#goog_game_inter